

MSSC 6250 – Statistical Machine Learning

Midterm Examination

Time: 75 minutes

Total: 100 points

Closed book. One formula sheet allowed.

Answer each question on the blank page immediately following it.

Name: _____

Student ID: _____

Problem 1 — Statistical Learning Framework (15 pts)

Suppose

$$Y = f(X) + \varepsilon, \quad \mathbb{E}(\varepsilon) = 0.$$

(a) (6 pts) Explain clearly:

- Prediction vs inference
- Reducible vs irreducible error

(b) (5 pts) Show that

$$\mathbb{E}[(Y - \hat{f}(x))^2] = [f(x) - \hat{f}(x)]^2 + \text{Var}(\varepsilon).$$

Explain the meaning of each term.

(c) (4 pts) Explain the bias–variance tradeoff.

- What happens to bias as model flexibility increases?
- What happens to variance?

Problem 2 — Linear Regression Interpretation (25 pts)

A department studies whether using AI tools (e.g., ChatGPT) is associated with exam performance.

Outcome: Y = Final exam score (out of 100)

Predictor: AIUse (1 = regularly uses AI tools, 0 = does not)

Model 1

$$Y = \beta_0 + \beta_1 \text{AIUse} + \varepsilon$$

Variable	Estimate	Std. Error	p-value
Intercept	78	1.5	< 0.001
AIUse	+6	2.0	0.003

Model 2

$$Y = \beta_0 + \beta_1 \text{AIUse} + \beta_2 \text{GPA} + \varepsilon$$

Variable	Estimate	Std. Error	p-value
Intercept	40	5	< 0.001
AIUse	−3	1.2	0.02
GPA	+12	1.5	< 0.001

- (a) Interpret the AIUse coefficient in Model 1.
- (b) Interpret the AIUse coefficient in Model 2. How does it differ from Model 1?
- (c) Explain why the coefficient changes from +6 to −3 after including GPA.
- (d) Which model provides a more reliable estimate of the association? Explain.

Problem 3 — Formulation of Classification Methods (20 pts)

Suppose $Y \in \{0, 1\}$ and $X \in \mathbb{R}^p$.

- (a) Logistic Regression: Write the model for $P(Y = 1 \mid X = x)$ in log-odds form and state how parameters are estimated.
- (b) Linear Discriminant Analysis (LDA): State the distributional assumption, covariance assumption, and quantities estimated.
- (c) Quadratic Discriminant Analysis (QDA): State the distributional assumption, how it differs from LDA, and the effect on the decision boundary.
- (d) Naive Bayes: Write the factorization of $p(X \mid Y = k)$ and explain the independence assumption.

Problem 4 — Resampling & Cross-Validation (20 pts)

- (a) Describe the K -fold cross-validation procedure.
- (b) Compare LOOCV and 10-fold CV in terms of bias and variance.
- (c) Why does training error underestimate test error?
- (d) Why is cross-validation useful for choosing tuning parameters?

Problem 5 — Model Selection & Regularization (20 pts)

- (a) Compare best subset selection and forward stepwise selection.
- (b) Why does least squares variance increase when $n \approx p$ or $n < p$?
- (c) Define ridge regression and explain the role of λ .
- (d) Compare ridge and lasso. Which performs variable selection and why?

